5.1 Angles of Triangles

Geometry 5.1 · Student sheet

Name _____ Period ____ Date _____

I can…

- classify triangles by their sides and by their angles.
- use the Triangle Sum Theorem to find angle measures.
- use the Exterior Angle Theorem to find angle measures.

Vocabulary · fill in the missing word

Scalene triangle — a triangle with no _____ sides

Isosceles triangle — a triangle with at least two _____ sides

Equilateral triangle — a triangle with three _____ sides

Exterior angle — an angle formed by one side of a polygon and the _____ of an adjacent side

Corollary — a _____ that can be proved easily using a theorem

Part 1 · Discovery: tear the corners

Tear off the corners of your triangle and line them up. Then measure on a second triangle with one side extended.

m∠1	m∠2	m∠3	Sum	Outside angle	Sum of the two far inside angles

The angles of a triangle add to _____. The outside angle equals _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Classifying triangles: by sides: scalene, isosceles, equilateral. By angles: acute, right, obtuse, or _____ (all angles congruent).

Example 1

Classify each triangle by its sides and by its angles.

a. angles 30°, 60°, 90°; no congruent sides

b. angles 40°, 40°, 100°; two congruent sides

Triangle Sum Theorem: the sum of the measures of the interior angles of a triangle is _____.

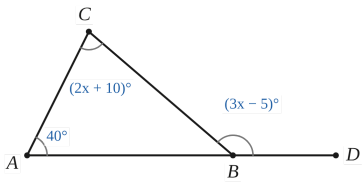
Example 2

Two angles of a triangle measure 52° and 71°. Find the third angle.

Exterior Angle Theorem: the measure of an exterior angle equals the _____ of the measures of the two nonadjacent interior angles.

Example 3

Find x and the measure of the exterior angle.



Example 4

The angles of a triangle are in the ratio 2 : 3 : 4. Find the three angles.

Corollary: the acute angles of a right triangle are _____.

Example 5

The acute angles of a right triangle measure (2x)° and (3x − 5)°. Find both angles.

Part 3 · Partner practice

Partner 1

Two angles of a triangle measure 35° and 85°. Find the third angle.

Partner 2

An exterior angle of a triangle measures 130°. One nonadjacent interior angle measures 72°. Find the other nonadjacent interior angle.